Andres Arreola

Mechanical Engineering
a.arreola22@yahoo.com | (916) 952-8463 | linkedin.com/in/anarreol

Education

California Polytechnic State University – San Luis Obispo (Cal Poly) June 2026
B.S. Mechanical Engineering | GPA: 3.44 | Dean's List (2x)
Relevant Coursework: Fluid Mechanics I & II, Thermodynamics I & II, Heat Transfer, Thermal System Design, Energy Conversion, Mechanical Systems Design, Measurements & Data Analysis, System Dynamics
Involvement: Water Power Club, ASHRAE Cal Poly, Society of Hispanic Professional Engineers

Engineering Experience

Hydropower Collegiate Competition – U.S. Department of Energy Aug 2025 – Present
Design Lead — 1**st Place Overall | 1st Place Build & Test | 1st Place Community Connections**

- Led a cross-functional team of 4 to design a 1:5 scale cross-flow (Banki) turbine model targeting ~133 kW full-scale shaft power at 73% efficiency using hydraulic similitude laws and EES thermodynamic modeling
- Conducted hydropower site feasibility analysis using SCADA pressure and flow data (4–31 cfs, 79–148 ft head) at a municipal pressure-reducing station to characterize annual energy potential
- Developed experimental test procedures using a Prony brake dynamometer, turbine flow meter, and pressure transducers; analyzed performance data to generate power and efficiency curves
- Built data reduction tools in Python (matplotlib) and Excel to produce publication-quality performance curves and scale model-to-prototype power projections
- Selected optimal turbine type via weighted decision matrix analyzing specific speed, head compatibility, and flow variability across five candidate turbine designs
- Engaged hydropower industry stakeholders and community partners to present design feasibility and address technical, regulatory, and economic challenges in small-scale hydropower deployment

Automotive Radiator Design – Thermal System Design Feb 2025 – Mar 2025

- Modeled a 2012 Ford Mustang cooling system in EES, calculating Vehicle Specific Power (73.3 kW) under extreme driving conditions (75 mph, 8.1% grade) by summing aerodynamic drag, rolling resistance, and grade force.
- Applied LMTD method to size a radiator heat exchanger for a 50/50 ethylene glycol-water coolant, determining the required heat transfer coefficient-area product and fluid mass flow rates
- Identified air-side convective resistance as the dominant thermal resistance, recommending increased fin density as the primary design improvement over coolant-side enhancements.

Owner / Operator – Landscaping Services, Self-Employed July 2018 – June 2024

- Managed all operations of a multi-client residential landscaping business including scheduling, bookkeeping, customer relations, and equipment maintenance
- Diagnosed and repaired mechanical equipment by interpreting manufacturer technical documentation; maintained continuous operational reliability across all contracted clients

Awards & Recognition

1st Place Overall – DOE Hydropower Collegiate Competition 2026
Top national finish across all judged categories: design report, build & test performance, and community connections
Cal Poly Scholars 2024 – Present
Merit program recognizing high-achieving undergraduate students from low-income backgrounds
Seal of Biliteracy – English & Spanish 2018
State-recognized achievement in English and Spanish linguistic proficiency

Technical Skills

CAD & Simulation: SolidWorks, SolidWorks FEA, SolidWorks Flow Simulation, AutoCAD
Programming & Analysis: MATLAB, Python (matplotlib, scipy), Simulink, EES, C++, Excel
Instrumentation: Pressure transducers, flow meters, thermocouples, dynamometers, data acquisition systems
Languages: English (native), Spanish (fluent)